Does Embodiment Help? Evaluating the Effect of a Corporeal Virtual Agent on Trust and Interaction Naturalness in a RAG-Based Document Consultation System

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***Abstract*—Document consultation systems based on Retrieval-Augmented Generation (RAG) typically present answers in plain text, which may constrain the naturalness and trust of the interaction. This paper presents Ariel, a web-native corporeal virtual agent (CVA) implemented with Three.js and @pixiv/three-vrm that acts as the front end of a RAG system for consulting document files in a document-management platform. Ariel communicates generated answers through synthesized speech and a 3D avatar with idle, listening and speaking animations, latency cues, and emotional expressions controlled by a large language model (LLM). We report a between-subjects expert pilot ($N = 8$) comparing the CVA (Condition A) against an equivalent text-only interface (Condition B) that shares an identical RAG and LLM back end, isolating the effect of embodiment. We measured perceived trust (Trust in Automation, TiA), interaction naturalness (Godspeed), usability (System Usability Scale, SUS) and factual retention. The CVA raised perceived trust (TiA trust 4.83 vs. 3.58) but, as implemented, was rated *lower* on anthropomorphism (1.60 vs. 2.50), animacy and likeability, while usability (SUS 73.8 vs. 72.5) and retention (2.25 vs. 2.50 of 3) were equivalent. Qualitative closing interviews attribute the naturalness penalty to synthetic voice and simulated lip sync rather than to embodiment itself. We conclude that the bottleneck is multimodal fidelity, not the mere presence of a body.**

***Index Terms*—Corporeal virtual agents, embodiment, RAG, perceived trust, HCI, VRM, document consultation.**

I. Introduction

The management of document databases, particularly those organized through archival techniques, requires consulting files that group multiple related documents. Retrieval-Augmented Generation (RAG) connects such collections with large language models (LLMs), enabling contextualized answers grounded in the real content of the file [1], [14], [15]. However, most RAG implementations present answers as plain text, much like a traditional chat interface.

Although functional, this presentation mode can feel limited when the goal is a consultation experience perceived as natural, close and trustworthy. Human–computer interaction (HCI) studies suggest that the presence of a corporeal virtual agent (CVA) can positively influence perceived trust, interaction naturalness and answer comprehension [7], [8], [10]. Yet specific evidence on CVAs applied to document RAG systems is scarce, and it is not obvious that adding a body is beneficial in a task whose value lies in factual precision.

This work investigates whether presenting RAG answers through a corporeal virtual agent—with synthesized voice, animations and emotional expressions—produces measurable differences in perceived trust and interaction naturalness compared to an equivalent text interface. We present Ariel, a CVA integrated in the RAG platform, and a comparative evaluation protocol designed to answer the following research questions:

- **RQ1:** What effect does using a CVA as the front end of a RAG system have on *perceived trust* in the generated answers, compared with a text-based document consultation interface?
- **RQ2:** What effect does the CVA have on the *perceived naturalness* of the interaction and on comprehension of answers in document consultation tasks, compared with a chat-style text interface?
- **RQ3:** What effect does the CVA have on *perceived usability* and on *information retention*, compared with a text interface?

Our contribution is threefold: (i) the design and implementation of Ariel as a web-native CVA built with Three.js and the Virtual Reality Model (VRM) avatar format, over a RAG back end; (ii) a controlled between-subjects protocol that isolates embodiment; and (iii) empirical evidence—quantitative and qualitative—showing a dissociation between trust (improved by embodiment) and naturalness (degraded by low multimodal fidelity). Section II reviews related work; Section III describes Ariel; Section IV the study design; Section V the results; Section VI the discussion; and Section VII the conclusions.

II. Related Work

Retrieval-Augmented Generation. RAG systems overcome a central limitation of LLMs: their dependence on static, internal knowledge. By integrating retrieval from external sources, RAG improves relevance, currency and grounding of answers [1], [14], [15], [16]. A key finding is that the agent must be connected to a verifiable document memory that reduces hallucinations and improves traceability [1], [14]; knowledge-grounded conversation further requires local context to stay coherent over long dialogues [16]. PersonaRAG shows that RAG improves when it incorporates user-centric agents that adapt retrieval and generation to needs in real time [2], a principle directly applicable to Ariel [3], [5]. Studies of user behavior with generative information-retrieval systems highlight how users formulate and verify queries [17], while knowledge-adaptive generative agents extend RAG toward up-to-date simulations [18].

Corporeal virtual agents and interaction. Studies in augmented and virtual reality show that an agent's visual presence can influence perceived intelligence, trust, closeness, usefulness and social presence [7], [8], [10], [13], [19]. However, the body must be designed with care: an agent that is overly anthropomorphic, unnatural or poorly synchronized can produce discomfort or loss of trust—the Uncanny Valley effect [24], [7]. Foundational work on embodied conversational agents establishes that combining natural language with non-verbal cues is fundamental to interactions perceived as natural [4], while research on relational trust shows that conversational agents designed as explanatory, orienting interfaces favor perceived trust [5]. Longitudinal studies of companion virtual agents further show that engagement and user attitudes evolve over repeated sessions [20].

Latency, multimodality and lip sync. Conversational naturalness depends on turn management, visual feedback and latency. Pauses, facial expressions and waiting cues mitigate the negative perception of latency in LLM-powered agents [9], [12]; allowing verbal interruptions improves perceived efficiency [6]. In browser speech-synthesis systems, lip sync quality has been shown to affect perceived naturalness and presence [8], [12].

Evaluation of CVAs. The LLM + RAG + multimodal-agent combination has outperformed purely textual, scripted variants in task success and perceived naturalness, especially in high-impact domains such as health, education and institutional services [11]. The most common instruments to evaluate CVAs are the Godspeed Questionnaire [22], the Trust in Automation scale (TiA) [21] and the System Usability Scale (SUS) [23], combined with information-retention tasks to measure effective comprehension.

III. System Design: Ariel

Ariel is a CVA integrated in a document-file management platform. It acts as a consultation interface over the documents of the active file, retrieving relevant information via RAG and communicating it through a 3D avatar with synthesized voice. The system has three components: avatar rendering (front end), the RAG pipeline (back end) and browser speech synthesis (Fig. 2); Fig. 1 shows the CVA condition in operation.

Front end. The avatar is rendered with **Three.js** and **@pixiv/three-vrm**, loading VRM avatars (a glTF-based humanoid-avatar format) directly in the browser, with no external game engine; this removes the need for a Unity build and

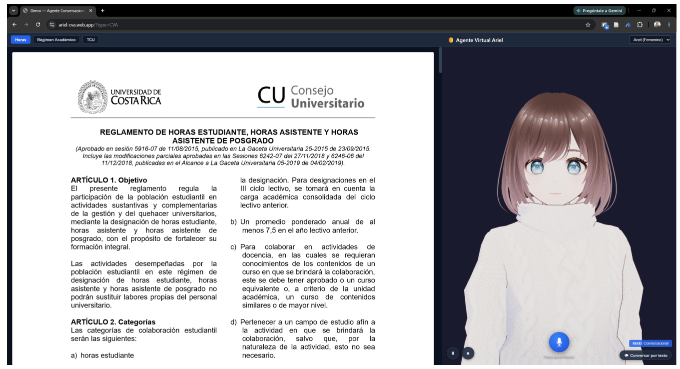

Figure 1. The CVA condition (Condition A) in operation: the 3D avatar (right) is rendered beside the document viewer showing the active file—a University of Costa Rica regulation (left).

improves reproducibility. The `AvatarController` manages dynamic VRM loading with runtime model swapping, a natural rest pose (arms lowered, head toward camera) applied from the default T-pose, and procedural idle animations (sinusoidal breathing, random blinking every 2–6 s, gentle head sway). Because the Web Speech API does not expose the audio stream, *simulated* lip sync is used: the `onstart`, `onboundary` and `onend` events of `SpeechSynthesisUtterance` drive the `Aa`/`Oh` blend shapes during speech. The LLM can invoke emotional expressions (happy, sad, neutral, surprised, relaxed, angry) mapped to VRM expressions with smooth interpolation.

Back end. A secure intermediary (Node/Express locally, deployed as Firebase Cloud Functions in production) processes each query in three stages: (1) *retrieval* via semantic search over the active file's vector store; (2) *generation*, integrating retrieved fragments as context for the LLM (OpenAI GPT-4.1-nano) together with behavior instructions; and (3) *response*, returning text plus an emotion-action metadatum that the controller interprets. **Speech and experimental toggle.** Audible narration uses the **Web Speech API**, removing external text-to-speech (TTS) dependencies and reducing latency; *thinking* animations cover retrieval and generation latency [9], [12]. A **"Conversational Mode" toggle** switches the system between the two experimental conditions without reloading, simultaneously controlling avatar visibility, grid layout, automatic TTS and speech-to-text (STT). In Condition B (text) the layout is `PDF 80% / Chat 20%`; in Condition A (CVA) it is `PDF 65% / Avatar 35%`, expanding to include a collapsible chat when requested. The retrieval and generation engine is *identical* in both conditions, so any observed difference is attributable to embodiment and its communicative modality.

IV. Methodology

Design. We ran a *between-subjects expert pilot* with two conditions: A (CVA: 3D avatar + synthesized voice + STT + animations) and B (text baseline: same platform with text chat only, no body or voice). An expert pilot was chosen because evaluation with real end users would require approval from the University of Costa Rica

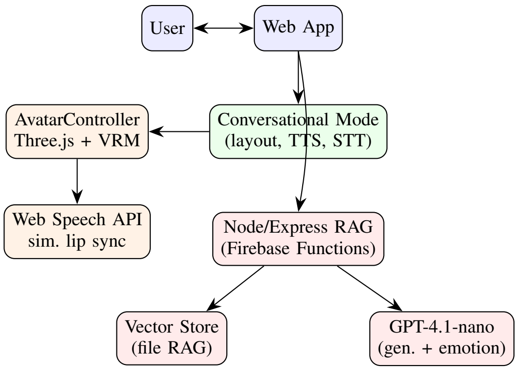

Figure 2. Ariel architecture. The RAG/LLM back end is shared by both conditions; the toggle enables or disables embodiment, voice and the avatar.

Scientific Ethics Committee, beyond the course time frame; experts in HCI or document systems provide structured evaluative feedback to refine the protocol. The between-subjects design avoids learning effects and direct-comparison bias.

Participants. Eight experts with academic or professional experience in HCI, information retrieval, conversational agents or document management, randomly assigned to condition ($n = 4$ per condition). One additional textual participant (TXT-01) was excluded for not completing all instruments, preserving a balanced $N = 8$. No personally identifiable data are reported.

Test file. The file comprises three public, formal regulations of the University of Costa Rica (Student/Assistant Hours; Student Academic Regime; University Community Work, TCU), selected because their normative nature is representative of the platform's files, their detail is unfamiliar to most participants (avoiding prior-knowledge bias), and they contain specific, verifiable data enabling objective retention questions.

Tasks and instruments. Each participant performed three tasks: T1, a verification query (factual answer, post-task TiA, RQ1); T2, a cross-regulation synthesis query integrating two regulations (post-task Godspeed, RQ2); and T3, three point-data queries with objectively verifiable answers (triple the student-hour wage; minimum passing grade 7.0; minimum 8 students per TCU project), feeding the retention questionnaire, with SUS administered post-session (RQ3). Instruments: TiA [21] (12 items, 7-point), Godspeed [22] (5-point semantic differential), SUS [23] (10 items, normalized 0–100) and an ad-hoc 3-item retention questionnaire, plus evaluator notes and a 5-question semi-structured closing interview (Nielsen-style structured observation [25]).

Procedure and analysis. Sessions (45–55 min) followed: consent, brief practice, T1–T3 with between-task instruments, SUS, and closing interview. Given the small sample, analysis is descriptive (means, standard deviations, trend comparison), complemented by qualitative interview coding. This pilot does not seek statistical significance but informs protocol refinement for a future study with real users.

Table I
QUANTITATIVE RESULTS BY CONDITION (MEAN ± SD, $N = 8$).

Metric	**A: CVA** (n=4)	**B: Text** (n=4)	**RQ**
TiA trust (1–7)	4.83 ± 2.01	3.58 ± 2.15	RQ1
TiA distrust (1–7)	2.15 ± 0.90	1.85 ± 0.38	RQ1
SUS (0–100)	73.8 ± 6.0	72.5 ± 14.9	RQ3
Retention (0–3)	2.25 ± 0.96	2.50 ± 1.00	RQ3

Table II
GODSPEED SUBSCALES (1–5 SEMANTIC DIFFERENTIAL).

Subscale	**A: CVA**	**B: Text**
Anthropomorphism	1.60 ± 0.85	2.50 ± 0.53
Animacy	2.00 ± 0.65	3.04 ± 0.75
Likeability	3.15 ± 0.57	3.85 ± 0.84
Perceived intelligence	3.80 ± 0.75	3.90 ± 0.26
Perceived safety	3.25 ± 0.50	4.00 ± 0.00

V. RESULTS

Table I summarizes the quantitative results by condition; the Godspeed subscales appear in Table II.

Perceived trust (TiA, RQ1). The CVA condition showed higher perceived trust on the positive trust items (mean 4.83 vs. 3.58 on a 7-point scale), with comparably low distrust (2.15 vs. 1.85). The trust response was, however, heterogeneous: one CVA expert rated trust low ($2/7$), reporting that the very speed of the answers raised suspicion. Textual participants anchored their trust on directly seeing the cited article in the PDF ("what made me trust it was seeing the article to verify").

Interaction naturalness (Godspeed, RQ2). Counter to the embodiment hypothesis, the CVA was rated *lower* than the text interface on anthropomorphism (1.60 vs. 2.50), animacy (2.00 vs. 3.04), likeability (3.15 vs. 3.85) and perceived safety (3.25 vs. 4.00). Perceived intelligence was essentially equal (3.80 vs. 3.90). Every CVA participant spontaneously flagged the synthetic voice and rigid movement as artificial ("it's not natural…evaluating the voice, it's not natural, the mouth doesn't move properly"; "more realistic, less of a manga cartoon").

Usability and retention (SUS, RQ3). SUS was equivalent and above the 68 acceptability threshold in both conditions (73.8 vs. 72.5). Retention was likewise equivalent and high (2.25 vs. 2.50 out of 3). Embodiment neither helped nor hurt usability or effective information transfer; both interfaces were described as fast, easy and intuitive.

VI. DISCUSSION

RQ1: embodiment raised perceived trust. On average the CVA produced higher trust (TiA 4.83 vs. 3.58) with similarly low distrust, consistent with literature reporting that visual presence and voice increase perceived trust and social presence [7], [8], [5]. The effect was not uniform: the dominant trust cue in *both* conditions was the verifiable source—textual users trusted because they could see the cited article, echoing the RAG principle that grounding in a verifiable memory underpins reliability [1], [14]. Embodiment therefore acts as a trust amplifier layered on top of source visibility, not as a substitute for it. A practical implication is that a CVA should make its sources explicit and visible, since immediacy alone can paradoxically breed suspicion (the “too-fast” answers reported by one expert).

RQ2: embodiment, as implemented, lowered perceived naturalness. The most salient—and counterintuitive—finding is that the embodied agent scored *below* the plain text interface on anthropomorphism, animacy and likeability. This is consistent with the Uncanny Valley [24]: a body with low-fidelity multimodal output (browser-synthesized voice, simulated lip sync decoupled from real audio, procedural-but-rigid motion, a stylized manga avatar) introduces artificiality cues that a familiar, frictionless text chat simply does not have. The qualitative data converge unanimously on voice and movement as the culprits, while perceived intelligence—driven by the shared RAG/LLM back end—remained equal across conditions. The lesson aligns with work on multimodal feedback and lip sync [8], [12], [4]: the presence of a body is not sufficient; its perceived naturalness is gated by the fidelity and synchronization of voice and motion.

RQ3: embodiment was usability- and retention-neutral. SUS was acceptable and statistically indistinguishable between conditions, and retention was high in both, indicating that the identical RAG core delivered the factual content equally well regardless of presentation. Embodiment did not impose a usability cost—an encouraging result—but also did not improve comprehension, in contrast with high-impact multimodal results elsewhere [11], plausibly because our task is short, factual and text-anchored (the PDF is always visible).

Synthesis. Embodiment improved trust, degraded perceived naturalness, and left usability and retention unchanged. The dissociation indicates that the bottleneck is *multimodal fidelity*, not embodiment per se: the same body that lent authority hurt naturalness because its voice and motion were synthetic. This reframes the design goal from “add an avatar” to “raise voice and animation fidelity to the level where embodiment’s trust benefit is not offset by an artificiality penalty.”

Limitations. The pilot is small ($N = 8$) and descriptive, without statistical power; experts are not the end-user population; the between-subjects design precludes within-subject comparison; the synthetic-voice/simulated-lip sync prototype likely represents a lower bound on achievable CVA naturalness; and the single-session, short-task setting limits ecological validity. The retention and TiA SDs are wide, so trends should be read as hypotheses for a larger study.

VII. Conclusions and Future Work

We presented Ariel, a web-native corporeal virtual agent (Three.js + VRM) over a RAG back end, and a controlled between-subjects pilot isolating embodiment in a document-consultation task. The evidence shows a clear dissociation: embodiment *increased perceived trust* (TiA 4.83 vs. 3.58) but *decreased perceived naturalness* (anthropomorphism 1.60 vs. 2.50), while leaving usability (SUS 73.8 vs. 72.5) and retention (2.25 vs. $2.50/3$) unchanged. Qualitative data attribute the naturalness penalty to low-fidelity voice and motion rather than to embodiment itself, indicating that multimodal fidelity—not the mere presence of a body—is the decisive design variable. Future work should replace browser TTS with high-fidelity neural voices and audio-driven lip sync, test more realistic avatars, and always surface verifiable sources to consolidate the trust benefit.

Publication projection. At present there is *no* plan to submit this study to the Scientific Ethics Committee or to pursue formal data collection for academic publication; the work is conducted as a course research exercise. Should the prototype’s multimodal fidelity be improved, a larger study with real users would be the natural next step toward publication.

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